



Dental Assistance App

1. Literature Review

1.1 Introduction

Access to dental care remains a significant challenge, especially for low-income populations who cannot afford private treatment. At the same time, dental students require real clinical cases to develop their practical skills. This creates an opportunity to bridge the gap between patients in need and students seeking hands-on experience through a structured digital platform.

1.2 Access to Dental Care

Oral health is a critical part of overall health; however, many individuals delay or avoid treatment due to financial limitations. Community dentistry studies show that underserved populations often suffer from untreated dental conditions due to lack of affordable services.

Free dental services are usually limited to university clinics or charity initiatives, which often face issues such as overcrowding, long waiting times, and inefficient case management.

1.3 Role of Technology in Healthcare

Digital health platforms have transformed how healthcare services are delivered. Web and mobile applications now enable patients to connect with healthcare providers efficiently.

Online systems help in:

- Managing appointments
- Reducing waiting time
- Improving communication
- Organizing patient data

However, most platforms focus on paid services and do not target patients who need free or low-cost care.

1.4 Existing Systems and Related Work

Existing healthcare platforms mainly focus on:

- **Booking appointments with professional doctors**
- **Telemedicine consultations**
- **Electronic health records**

On the other hand, educational systems for medical students often rely on simulations rather than real patient interaction.

Although charity dental programs exist, they lack digital systems that efficiently connect patients with dental students and manage case workflows.

1.5 Research Gap

There is a clear lack of systems that:

- **Connect dental students with real patients**
- **Provide affordable or free dental care**
- **Manage case requests and sessions digitally**

Most current solutions either focus on professional services or education separately, without integrating both aspects.

1.6 Proposed Solution: DentLink

DentLink is proposed as a web-based platform that connects dental students with patients requiring free or low-cost dental care.

The system aims to:

- **Improve access to dental services for underserved patients**
- **Provide real clinical experience for students**
- **Enable efficient case management and communication**
- **Ensure safety through admin approval and verification**

2. Project Planning & Management

2.1 Project Overview

DentLink is a web-based dental assistance system designed to connect patients with dental students. The platform allows patients to submit cases, while students can browse, request, and manage treatment sessions under supervision.

2.2 Objectives

The main objectives of the project are:

- **To provide accessible dental care for patients**
 - **To support dental students with practical training**
 - **To create a structured system for managing cases and sessions**
 - **To improve efficiency using digital solutions**
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2.3 Stakeholders

The key stakeholders include:

- **Patients (service beneficiaries)**
 - **Dental students (service providers)**
 - **Admin (system manager and verifier)**
 - **Educational institutions**
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2.4 Project Scope

In Scope:

- **User registration (patient and doctor)**
- **Doctor verification via ID and admin approval**
- **Case creation and management**
- **Case request system**
- **Session tracking and attendance confirmation**
- **Transportation cost handling**

Out of Scope:

- **Automated medical diagnosis**
 - **Online payment processing**
 - **Emergency medical services**
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2.5 Work Breakdown Structure (WBS)

The project is divided into the following phases:

1. Requirements Analysis

- **Identify system requirements**
- **Analyze user needs**

2. System Design

- **ER Diagram**
- **Database design**
- **Use case diagram**
- **UI/UX design**

3. Development

- **Frontend implementation**
- **Backend development**

- **Database integration**

4. Testing

- **Unit testing**
- **System testing**
- **User acceptance testing**

5. Deployment

- **System launch**
- **Monitoring and maintenance**

2.6 Project Timeline

Phase	Duration
Requirements Analysis	1 week
System Design	2 weeks
Development	4 weeks
Testing	2 weeks
Deployment	1 week

2.7 Risk Management

Risk	Mitigation Strategy
Doctor not verified properly	Admin approval system
Patients missing appointments	Use reminders / confirmation (OTP/QR)
Low number of cases	Promote platform awareness
Data privacy issues	Secure authentication and data protection
System misuse	Verification and monitoring

2.8 Tools and Technologies

- **Frontend: HTML, CSS, JavaScript**
 - **Backend: .Net**
 - **Database: SQL Server**
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2.9 Development Methodology

The Agile (Scrum) methodology will be used to develop the system. This allows:

- **Iterative development**
 - **Continuous feedback**
 - **Flexibility in handling changes**
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2.10 Deliverables

The final deliverables include:

- **Fully functional web application**
 - **Database system**
 - **System documentation**
 - **Final project report**
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3. Requirements

3.1 User Registration (Sign Up)

- The system shall provide free user registration.
 - The system shall allow users to select their role:
 - Patient
 - Doctor (Dental Student)
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3.2 Patient Profile Page

- The system shall allow patients to enter the following information:
 - Full name
 - Phone number
 - Age
 - Residential area (for distance and transportation calculation)

Enhancement:

- The system may support automatic location detection using GPS instead of manual input.
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3.3 Doctor Profile Page

- The system shall allow doctors to enter the following information:
 - Full name
 - University / Faculty
 - Academic year
 - Student ID card (for verification)

Important Requirement:

- The system shall require **admin approval** before a doctor can access or accept cases.

3.4 Case Submission (Patient)

- The system shall allow patients to submit a dental case request including:
 - Free-text description of the condition
 - Selection of case type:
 - Filling
 - Extraction
 - Cleaning
 - Surgery
 - Uploading an image (optional)

Advanced Enhancement:

- The system may include an intelligent chatbot that:
 - Asks guided questions (e.g., duration of pain, swelling, bleeding)
 - Helps classify the case automatically

3.5 Case Listing (Doctor)

- The system shall display available cases to doctors, including:
 - Case type
 - Patient location
 - Distance (e.g., 5 km)

Enhancements:

- The system shall allow sorting cases by:
 - Nearest distance
 - Most critical cases
 - Most recent cases

3.6 Attendance Confirmation and Session Start

- The system shall allow confirmation when the patient arrives at the clinic.
- The doctor shall be able to mark the patient as "arrived" and start the session.

Important Requirement:

- The system shall support secure confirmation using:
 - QR Code
 - OTP (One-Time Password)
(to prevent misuse or fraud)

4.Functional Requirements

The DentLink system shall provide the following functionalities:

4.1 User Registration and Authentication

- The system shall allow users to sign up as either a **patient** or a **doctor (dental student)**.
- The system shall allow users to log in securely using their credentials.
- The system shall allow doctors to upload their **student ID card** for verification.
- The system shall allow the admin to **approve or reject doctor accounts**.
- The system shall restrict unapproved doctors from accessing system functionalities.

4.2 Patient Features

- The system shall allow patients to create and manage their profiles (name, age, address, phone).
- The system shall allow patients to create a new **dental case**.
- The system shall allow patients to:
 - Enter a description of their condition
 - Select the case type (e.g., filling, extraction, cleaning, surgery)

- Upload an image (optional)
 - The system shall allow patients to view incoming **doctor requests**.
 - The system shall allow patients to **accept or reject case requests**.
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4.3 Doctor Features

- The system shall allow doctors to browse available cases.
 - The system shall allow doctors to view case details (type, description, location, image).
 - The system shall allow doctors to send a **case request** to a patient.
 - The system shall allow doctors to include a **transportation cost** in the request.
 - The system shall allow doctors to start and complete treatment sessions after approval.
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4.4 Case Request Management

- The system shall create a **case request record** when a doctor applies for a case.
 - The system shall store request details including:
 - doctor ID
 - case ID
 - request date
 - status (pending, accepted, rejected)
 - transportation cost
 - The system shall update the request status based on patient response.
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4.5 Session Management

- The system shall create a session when a case request is accepted.
- The system shall allow doctors to:
 - Start a session

- Mark patient arrival
 - End the session
 - The system shall store session details (start time, end time, attendance status).
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4.6 Admin Features

- The system shall allow the admin to view all registered doctors.
 - The system shall allow the admin to approve or reject doctors.
 - The system shall allow the admin to monitor system activity.
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4.7 Notifications and Confirmation

- The system shall notify patients when a doctor sends a request.
 - The system shall notify doctors when their request is accepted or rejected.
 - The system shall support confirmation mechanisms (e.g., OTP or QR code) for session validation.
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5.Non-Functional Requirements

The DentLink system shall satisfy the following quality attributes:

5.1 Performance

- The system shall support multiple users simultaneously without significant delays.
 - The system shall respond to user requests within an acceptable time (≤ 3 seconds).
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5.2 Security

- The system shall ensure secure authentication for all users.
- The system shall protect sensitive user data (e.g., personal information, images).
- The system shall restrict access based on user roles (patient, doctor, admin).

5.3 Usability

- The system shall provide a simple and user-friendly interface.
- The system shall be easy to use for non-technical users.
- The system shall minimize the number of steps required to complete tasks.

5.4 Availability

- The system shall be available 24/7 with minimal downtime.

5.5 Scalability

- The system shall be designed to handle increasing numbers of users and cases.

5.6 Reliability

- The system shall ensure data consistency and prevent data loss.

5.7 Maintainability

- The system shall be modular to allow easy updates and modifications.
- The codebase shall be well-structured and documented.

5.8 Compatibility

- The system shall be accessible عبر مختلف المتصفحات (Chrome, Edge, Firefox).
- The system shall support mobile and desktop devices.